

Module 02 Assignments

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Group 7.4

Question 5

Write out the form of the partial fraction decomposition of the function (as in Example 7). Do not determine the numerical values of the coefficients.

a.

$$\frac{x^5 + 1}{(x^2 - x)(x^4 + 2x^2 + 1)}$$

Compare polynomial degrees.

$$\deg(\text{numerator}) = 5 \quad \deg(\text{denominator}) = 6$$

$$\deg(\text{numerator}) < \deg(\text{denominator}) \implies \text{Decompose directly}$$

Factor the denominator.

$$x(x - 1)(x^2 + 1)(x^2 + 1)$$

Set up the decomposition.

$$\boxed{\frac{A}{x} + \frac{B}{x - 1} + \frac{Cx + D}{x^2 + 1} + \frac{Ex + F}{(x^2 + 1)^2}}$$

b.

$$\frac{x^2}{x^2 + x - 6}$$

Compare the polynomial degrees.

$$\deg(\text{numerator}) = 2 \quad \deg(\text{denominator}) = 2$$

$$\deg(\text{numerator}) \geq \deg(\text{denominator}) \implies \text{Perform polynomial long division first}$$

Rewrite the missing terms in the numerator

$$\frac{x^2 + 0x + 0}{x^2 + x - 6}$$

Divide the leading terms.

$$x^2 \div x^2 = 1$$

Put 1 in the quotient, then multiply it by the entire divisor

$$1(x^2 + x - 6) = x^2 + x - 6$$

Subtract

$$(x^2 + 0x + 0) - (x^2 + x - 6) = x^2 - x^2 + 0x - x + 0 - (-6)$$

$$= -x + 6$$

We stop dividing because the remainder $-x + 6$ has a degree 1, which is less than the divisor's degree 2.

Factor the denominator.

$$x^2 + x - 6 = (x + 3)(x - 2)$$

Therefore

$$1 + \frac{-x + 6}{(x + 3)(x - 2)}$$

Set up the partial fraction decomposition.

$$\boxed{1 + \frac{A}{x + 3} + \frac{B}{x - 2}}$$

Question 10

Evaluate the integral.

$$\int \frac{y}{(y+4)(2y-1)} dy$$

Set up partial fraction decomposition

$$\frac{y}{(y+4)(2y-1)} = \frac{A}{(y+4)} + \frac{B}{(2y-1)}$$

Multiply through by $(y+4)(2y-1)$

$$y = A(2y-1) + B(y+4)$$

Let $y = -4$

$$-4 = A(2(-4) - 1)$$

$$A = \frac{4}{9}$$

Let $y = \frac{1}{2}$

$$\frac{1}{2} = A\left(2 \cdot \frac{1}{2} - 1\right) + B\left(\frac{1}{2} + 4\right)$$

$$\frac{1}{2} = B\left(\frac{9}{2}\right)$$

$$B = \frac{1}{9}$$

Therefore

$$\frac{y}{(y+4)(2y-1)} = \frac{\frac{4}{9}}{(y+4)} + \frac{\frac{1}{9}}{(2y-1)}$$

Integrate

$$\int \left(\frac{4}{9(y+4)} + \frac{1}{9(2y-1)} \right) dy$$

$$\frac{4}{9} \ln |y+4| + \frac{1}{9} \left(\frac{1}{2} \ln |2y-1| \right)$$

$$\boxed{\frac{4}{9} \ln |y + 4| + \frac{1}{18} \ln |2y - 1| + C}$$

Question 12

Evaluate the integral.

$$\int_0^1 \frac{x - 4}{x^2 - 5x + 6} dx$$

Factor the denominator

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

$$\int_0^1 \frac{x - 4}{(x - 2)(x - 3)} dx$$

Set up partial fraction decomposition

$$\frac{x - 4}{(x - 2)(x - 3)} = \frac{A}{x - 2} + \frac{B}{x - 3}$$

Multiply through by $(x - 2)(x - 3)$

$$x - 4 = A(x - 3) + B(x - 2)$$

Let $x = 3$

$$3 - 4 = A(3 - 3) + B(3 - 2)$$

$$B = -1$$

Let $x = 2$

$$2 - 4 = A(2 - 3) + B(2 - 2)$$

$$A = 2$$

Therefore

$$\frac{x-4}{(x-2)(x-3)} = \frac{2}{x-2} - \frac{1}{x-3}$$

Integrate

$$\int_0^1 \frac{2}{x-2} - \frac{1}{x-3} dx$$

$$[2 \ln |x-2| - \ln |x-3|]_0^1$$

Evaluate the bounds

At $x = 1$

$$2 \ln |1-2| - \ln |1-3| = 2 \ln 1 - \ln 2$$

At $x = 0$

$$2 \ln |0-2| - \ln |0-3| = 2 \ln 2 - \ln 3$$

$$2 \ln 1 - \ln 2 - (2 \ln 2 - \ln 3)$$

$$\boxed{2 \ln 1 - \ln 2 - 2 \ln 2 + \ln 3}$$

Question 15

Evaluate the integral.

$$\int \frac{x^2}{x-1} dx$$

Perform polynomial long division

$$\begin{array}{r} x+1 \\ x-1 \overline{) x^2+0x+0} \\ \underline{x^2-x} \\ x+0 \\ \underline{x-1} \\ 1 \end{array}$$

Integrate

$$\int \left(x + 1 + \frac{1}{x-1} \right) dx$$

$$\boxed{\frac{1}{2}x^2 + x + \ln|x-1| + C}$$

Question 18

Evaluate the integral.

$$\int_1^2 \frac{3x^2 + 6x + 2}{x^2 + 3x + 2} dx$$

Perform polynomial long division

$$\begin{array}{r} 3 \\ x^2 + 3x + 2 \overline{) 3x^2 + 6x + 2} \\ \underline{3x^2 + 9x + 6} \\ -3x - 4 \end{array}$$

Since the remainder has degree 1 and the denominator has degree 2, stop.

Therefore

$$\frac{3x^2 + 6x + 2}{x^2 + 3x + 2} = 3 - \frac{3x + 4}{x^2 + 3x + 2}$$

Factor the denominator

$$x^2 + 3x + 2 = (x + 1)(x + 2)$$

So

$$\frac{-3x - 4}{(x + 1)(x + 2)} = \frac{A}{x + 1} + \frac{B}{x + 2}$$

Multiply through by $(x + 1)(x + 2)$

$$-3x - 4 = A(x + 2) + B(x + 1)$$

Let $x = -2$

$$-3(-2) - 4 = A(-2 + 2) + B(-2 + 1)$$

$$B = -2$$

Let $x = -1$

$$-3(-1) - 4 = A(-1 + 2) + B(-1 + 1)$$

$$A = -1$$

Therefore

$$\frac{-3x - 4}{(x + 1)(x + 2)} = -\frac{1}{x + 1} - \frac{2}{x + 2}$$

Integrate

$$\int_1^2 3 - \frac{1}{x + 1} - \frac{2}{x + 2} dx$$

$$[3x - \ln|x + 1| - 2\ln|x + 2|]_1^2$$

At $x = 2$

$$3(2) - \ln|2 + 1| - 2\ln|2 + 2| = 6 - \ln 3 - 2\ln 4$$

At $x = 1$

$$3(1) - \ln|1 + 1| - 2\ln|1 + 2| = 3 - \ln 2 - 2\ln 3$$

$$6 - \ln 3 - 2\ln 4 - (3 - \ln 2 - 2\ln 3)$$

$$\boxed{3 + \ln 2 + \ln 3 - 2\ln 4}$$

Question 19

Evaluate the integral.

$$\int_0^1 \frac{x^2 + x + 1}{(x+1)^2(x+2)} dx$$

Set up the partial fraction decomposition

Since the denominator contains the repeated linear factor $(x+1)^2$

$$\frac{x^2 + x + 1}{(x+1)^2(x+2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2}$$

Multiply through by $(x+1)^2(x+2)$

$$x^2 + x + 1 = A(x+1)(x+2) + B(x+2) + C(x+1)^2$$

Solve for the coefficients

Let $x = -2$

$$(-2)^2 + (-2) + 1 = A(-2+1)(-2+2) + B(-2+2) + C(-2+1)^2$$

$$4 - 2 + 1 = 0 + 0 + C$$

$$C = 3$$

Let $x = -1$

$$(-1)^2 + (-1) + 1 = A(-1+1)(-1+2) + B(-1+2) + C(-1+1)^2$$

$$1 - 1 + 1 = 0 + B + 0$$

$$B = 1$$

Let $x = 0$

$$(0)^2 + 0 + 1 = A(0+1)(0+2) + (1)(0+2) + (3)(0+1)^2$$

$$1 = 2A + 5$$

$$A = -2$$

Therefore

$$\frac{x^2 + x + 1}{(x + 1)^2(x + 2)} = -\frac{2}{x + 1} + \frac{1}{(x + 1)^2} + \frac{3}{x + 2}$$

Integrate

$$\int_0^1 \left(-\frac{2}{x + 1} + \frac{1}{(x + 1)^2} + \frac{3}{x + 2} \right) dx$$

$$\left[-2 \ln |x + 1| - \frac{1}{x + 1} + 3 \ln |x + 2| \right]_0^1$$

Evaluate the bounds

Upper bound $x = 1$

$$-2 \ln |1 + 1| - \frac{1}{1 + 1} + 3 \ln |1 + 2| = 3 \ln 3 - 2 \ln 2 - \frac{1}{2}$$

Lower bound $x = 0$

$$-2 \ln |0 + 1| - \frac{1}{0 + 1} + 3 \ln |0 + 2| = 3 \ln 2 - 2 \ln 1 - 1$$

Subtract

$$3 \ln 3 - 2 \ln 2 - \frac{1}{2} - (3 \ln 2 - 2 \ln 1 - 1)$$

$$\boxed{\frac{1}{2} - 5 \ln 2 + 3 \ln 3}$$

Question 21

Evaluate the integral.

$$\int \frac{dt}{(t^2 - 1)^2}$$

Factor the denominator

$$(t^2 - 1)^2 = (t - 1)^2(t + 1)^2$$

Set up the partial fraction

$$\frac{dt}{(t-1)^2(t+1)^2} = \frac{A}{t-1} + \frac{B}{(t-1)^2} + \frac{C}{t+1} + \frac{D}{(t+1)^2}$$

Factor through by $(t-1)^2(t+1)^2$

$$1 = A(t-1)(t+1)^2 + B(t+1)^2 + C(t-1)^2(t+1) + D(t-1)^2$$

Solve for the coefficients

Let $x = 1$

$$1 = A(1-1)(1+1)^2 + B(1+1)^2 + C(1-1)^2(1+1) + D(1-1)^2$$

$$1 = 0 + B(2)^2 + 0 + 0$$

$$1 = 4B$$

$$B = \frac{1}{4}$$

Let $x = -1$

$$1 = A(-1-1)(-1+1)^2 + B(-1+1)^2 + C(-1-1)^2(-1+1) + D(-1-1)^2$$

$$1 = 0 + 0 + 0 + D(-2)^2$$

$$1 = 4D$$

$$D = \frac{1}{4}$$

Let $x = 0$

$$1 = A(0-1)(0+1)^2 + \frac{1}{4}(0+1)^2 + C(0-1)^2(0+1) + \frac{1}{4}(0-1)^2$$

$$1 = -A + \frac{1}{4} + C + \frac{1}{4}$$

$$A - C = -\frac{1}{2}$$

Let $x = 2$

$$1 = A(2-1)(2+1)^2 + \frac{1}{4}(2+1)^2 + C(2-1)^2(2+1) + \frac{1}{4}(2-1)^2$$

$$1 = 9A + \frac{9}{4} + 3C + \frac{1}{4}$$

$$1 - \frac{9}{4} - \frac{1}{4} = 9A + 3C$$

$$-\frac{3}{2} = 9A + 3C$$

$$-3A - C = \frac{1}{2}$$

Solve the two equations

Subtract

$$A - C - (-3A - C) = -\frac{1}{2} - \left(\frac{1}{2}\right)$$

$$4A = -1$$

$$A = -\frac{1}{4}$$

Then

$$C = \frac{1}{4}$$

Therefore

$$\frac{dt}{(t-1)^2(t+1)^2} = -\frac{1}{4(t-1)} + \frac{1}{4(t-1)^2} + \frac{1}{4(t+1)} + \frac{1}{4(t+1)^2}$$

Integrate

$$\int \left(\frac{dt}{(t-1)^2(t+1)^2} = -\frac{1}{4(t-1)} + \frac{1}{4(t-1)^2} + \frac{1}{4(t+1)} + \frac{1}{4(t+1)^2} \right)$$

$$\boxed{\frac{1}{4} \left[\ln|t+1| - \frac{1}{t+1} - \ln|t-1| - \frac{1}{t-1} \right] + C}$$

Question 23

Evaluate the integral.

$$\int \frac{10}{(x-1)(x^2+9)} dx$$

Setup the partial fraction decomposition

$$\frac{10}{(x-1)(x^2+9)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+9}$$

Multiply through by $(x-1)(x^2+9)$

$$10 = A(x^2+9) + (Bx+C)(x-1)$$

Expand

$$10 = Ax^2 + A9 + Bx^2 - (B-C)x - C$$

Group like terms

$$(A+B)x^2 - (B-C)x + (A9-C)$$

Compare coefficients

$$A+B=0$$

$$B-C=0$$

$$A9-C=10$$

From

$$A+B=0$$

$$B=-A$$

From

$$B-C=0$$

$$-C = A$$

Substitute into

$$A9 + A = 10$$

$$10A = 10$$

$$A = 1$$

Therefore

$$B = -1 \quad C = -1$$

So

$$\frac{10}{(x-1)(x^2+9)} = \frac{1}{x-1} + \frac{-x-1}{x^2+9}$$

Integrate

$$\int \left(\frac{1}{x-1} - \frac{x}{x^2+9} - \frac{1}{x^2+9} \right) dx$$

First term

$$\int \frac{1}{x-1} dx = \ln |x-1|$$

Second term

$$- \int \frac{x}{x^2+9} dx$$

Let

$$u = x^2 + 9 \quad du = 2x dx$$

Then

$$x dx = \frac{1}{2} du$$

So

$$-\frac{1}{2} \int \frac{1}{u} du = -\frac{1}{2} \ln |u|$$

Substitute back

$$-\frac{1}{2} \ln(x^2 + 9)$$

Third term

$$\begin{aligned} \int \frac{1}{x^2 + 9} &= -\left(\frac{1}{3} \arctan\left(\frac{x}{3}\right)\right) \\ &= -\frac{1}{3} \arctan\left(\frac{x}{3}\right) \end{aligned}$$

Therefore

$$\boxed{\ln |x - 1| - \frac{1}{2} \ln(x^2 + 9) - \frac{1}{3} \arctan\left(\frac{x}{3}\right) + C}$$

Question 24

Evaluate the integral.

$$\int \frac{3x^2 - x + 8}{x^3 + 4x} dx$$

Factor the denominator

$$x^3 + 4x = x(x^2 + 4)$$

So

$$\int \frac{3x^2 - x + 8}{x(x^2 + 4)} dx$$

Set up the partial fraction decomposition

$$\frac{3x^2 - x + 8}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

Multiply through by $x(x^2 + 4)$

$$3x^2 - x + 8 = A(x^2 + 4) + (Bx + C)x$$

Expand

$$3x^2 - x + 8 = Ax^2 + 4A + Bx^2 + Cx$$

Group like terms

$$3x^2 - x + 8 = (A + B)x^2 + Cx + 4A$$

Compare coefficients

$$A + B = 3$$

$$C = -1$$

$$4A = 8$$

So

$$A = 2 \quad B = 1 \quad C = -1$$

Therefore

$$\frac{3x^2 - x + 8}{x(x^2 + 4)} = \frac{2}{x} + \frac{x - 1}{x^2 + 4}$$

Split the second numerator

$$\frac{x - 1}{x^2 + 4} = \frac{x}{x^2 + 4} - \frac{1}{x^2 + 4}$$

Integrate

$$\int \frac{2}{x} dx + \int \frac{x}{x^2 + 4} dx - \int \frac{1}{x^2 + 4} dx$$

First term

$$\int \frac{2}{x} dx = 2 \ln |x|$$

Second term

Let

$$u = x^2 + 4 \quad du = 2x \, dx$$

Then

$$x \, dx = \frac{1}{2} \, du$$

So

$$\begin{aligned} \int \frac{x}{x^2 + 4} \, dx &= \frac{1}{2} \int \frac{1}{u} \, du \\ &= \frac{1}{2} \ln |u| \\ &= \frac{1}{2} \ln(x^2 + 4) \end{aligned}$$

Third term

$$- \int \frac{1}{x^2 + 4} \, dx = -\frac{1}{2} \arctan\left(\frac{x}{2}\right)$$

Therefore

$$\boxed{2 \ln |x| + \frac{1}{2} \ln(x^2 + 4) - \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C}$$

Question 41

Make a substitution to express the integrand as a rational function and then evaluate the integral.

$$\int \frac{dx}{x\sqrt{x-1}}$$

Apply substitution

Let

$$u = \sqrt{x-1}$$

Square both sides

$$u^2 = x - 1$$

So

$$x = u^2 + 1$$

Differentiate

$$dx = 2u \, du$$

Rewrite the integral in terms of u

$$\int \frac{2u \, du}{(u^2 + 1)u}$$

Cancel u

$$\int \frac{2}{u^2 + 1} \, du$$

Recognize the arctan form

$$2 (\arctan (u)) = 2 \arctan (u) + C$$

Substitute back

$$\boxed{2 \arctan (\sqrt{x-1}) + C}$$

Question 42

Make a substitution to express the integrand as a rational function and then evaluate the integral.

$$\int \frac{dx}{2\sqrt{x+3}+x}$$

Apply substitution

Let

$$u = \sqrt{x+3}$$

Then

$$u^2 = x + 3$$

so

$$x = u^2 - 3$$

Differentiate

$$dx = 2u \, du$$

Substitute in the integral

$$\int \frac{2u \, du}{2u + u^2 - 3}$$

Factor the denominator

$$u^2 + 2u - 3 = (u + 3)(u - 1)$$

So

$$\int \frac{2u}{(u + 3)(u - 1)} \, du$$

Set up the partial fraction

$$\frac{2u}{(u + 3)(u - 1)} = \frac{A}{u + 3} + \frac{B}{u - 1}$$

Multiply through by $(u + 3)(u - 1)$

$$2u = A(u - 1) + B(u + 3)$$

Let $u = 1$

$$2(1) = A(1 - 1) + B(1 + 3)$$

$$B = \frac{1}{2}$$

Let $u = -3$

$$2(-3) = A(-3 - 1) + B(-3 + 3)$$

$$A = \frac{3}{2}$$

Therefore

$$\frac{2u}{(u+3)(u-1)} = \frac{\frac{3}{2}}{u+3} + \frac{\frac{1}{2}}{u-1}$$

Integrate

$$\begin{aligned} & \frac{3}{2} \int \frac{1}{u+3} du + \frac{1}{2} \int \frac{1}{u-1} du \\ &= \frac{3}{2} \ln|u+3| + \frac{1}{2} \ln|u-1| + C \end{aligned}$$

Substitute back

Since

$$u = \sqrt{x+3}$$

$$\boxed{\frac{1}{2} \left[3 \ln|\sqrt{x+3}+3| + \ln|\sqrt{x+3}-1| \right] + C}$$

Question 44

Make a substitution to express the integrand as a rational function and then evaluate the integral.

$$\int_0^1 \frac{1}{1+\sqrt[3]{x}} dx$$

Apply substitution

Let

$$u = \sqrt[3]{x}$$

Then

$$u^3 = x$$

and

$$dx = 3u^2 du$$

Convert the bounds

$$x = 0 \implies u = 0$$

$$x = 1 \implies u = 1$$

Rewrite the integral in terms of u

$$\int_0^1 \frac{3u^2}{u+1} du$$

Perform polynomial long division

$$\begin{array}{r} 3u - 3 \\ u + 1 \overline{) 3u^2 + 0u + 0} \\ \underline{3u^2 + 3u} \\ -3u + 0 \\ \underline{-3u - 3} \\ 3 \end{array}$$

Therefore

$$\frac{3u^2}{u+1} = 3u - 3 + \frac{3}{u+1}$$

So

$$\int_0^1 \left(3u - 3 + \frac{3}{u+1} \right) du$$

Integrate

$$= \left[\frac{3}{2}u^2 - 3u + 3 \ln |u+1| \right]_0^1$$

Evaluate at the bounds

Upper bound

$$\frac{3}{2}(1)^2 - 3(1) + 3 \ln |1+1|$$

$$3 \ln |2| - \frac{3}{2}$$

Lower bound

$$\frac{3}{2}(0)^2 - 3(0) + 3 \ln |0+1|$$

$$= 0$$

Subtract

$$\left(3 \ln |2| - \frac{3}{2}\right) - (0)$$

$$\boxed{3 \ln |2| - \frac{3}{2}}$$

Group 7.5

Question 1

Three integrals are given that, although they look similar, may require different techniques of integration. Evaluate the integrals.

a. s -Substitution

$$\int \frac{x}{1+x^2} dx$$

Let

$$u = x^2 + 1$$

Then

$$du = 2x dx \quad x dx = \frac{1}{2} du$$

Substitute

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \int \frac{1}{2} du$$

Integrate

$$\frac{1}{2} \ln |u| + C$$

Substitute back

$$\boxed{\frac{1}{2} \ln(x^2 + 1) + C}$$

b. Arctangent Form

$$\int \frac{1}{1+x^2} dx$$

$$\boxed{\arctan(x) + C}$$

c. Partial Fraction Decomposition

$$\int \frac{1}{1-x^2} dx$$

Factor the denominator

$$1-x^2 = (1-x)(1+x)$$

Set up the decomposition

$$\frac{1}{(1-x)(1+x)} = \frac{A}{1-x} + \frac{B}{1+x}$$

Multiply through by $(1-x)(1+x)$

$$1 = A(1+x) + B(1-x)$$

Let $x = 1$

$$1 = A(1+1) + B(1-1)$$

$$A = \frac{1}{2}$$

Let $x = -1$

$$1 = A(1-1) + B(1-(-1))$$

$$B = \frac{1}{2}$$

Therefore

$$\frac{1}{(1-x)(1+x)} = \frac{1}{2(1-x)} + \frac{1}{2(1+x)}$$

Integrate

$$\frac{1}{2} \int \frac{1}{1-x} dx + \frac{1}{2} \int \frac{1}{1+x} dx$$

$$\boxed{\frac{1}{2} [\ln |1+x| - \ln |1-x|] + C}$$

Question 3

Three integrals are given that, although they look similar, may require different techniques of integration. Evaluate the integrals.

a. u -Substitution

$$\int \frac{\ln x}{x} dx$$

Let

$$u = \ln(x) \quad du = \frac{1}{x} dx$$

Substitute

$$\int u du$$

$$\frac{u^2}{2}$$

Substitute back

$$\boxed{\frac{1}{2} \ln(x)^2 + C}$$

b. Integration by parts

$$\int \ln(2x) dx$$

Let

$$u = \ln(2x) \quad dv = dx$$

and

$$du = \frac{1}{x} dx \quad v = x$$

Apply integration by parts

$$\int u \, dv = uv - \int v \, du$$

$$\int x \ln x \, dx = x \ln(2x) - \int x \cdot \frac{1}{x} \, dx$$

$$\boxed{x \ln(2x) - x + C}$$

c. Integration by parts

$$\int x \ln x \, dx$$

Let

$$u = \ln x \quad dv = x \, dx$$

and

$$du = \frac{1}{x} \, dx \quad v = \frac{x^2}{2}$$

Apply integration by parts

$$\int x \ln x \, dx = \frac{1}{2} x^2 \ln x - \int \frac{x^2}{2} \left(\frac{1}{x} \right)$$

Simplify

$$\frac{1}{2} x^2 \ln x - \int \frac{x}{2}$$

Integrate

$$\boxed{\frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + C}$$

Question 4

Three integrals are given that, although they look similar, may require different techniques of integration. Evaluate the integrals.

a. Half-angle identity

$$\int \sin^2(x) dx$$

Use the half-angle identity

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

Substitute

$$\int \frac{1 - \cos(2x)}{2} dx$$

Split the integral

$$\frac{1}{2} \int 1 dx - \frac{1}{2} \int \cos(2x) dx$$

Integrate

$$\frac{x}{2} - \frac{1}{2} \left(\frac{1}{2} \sin(2x) \right) + C$$

$$\frac{x}{2} - \frac{1}{4} \sin(2x) + C$$

b. Odd power of sine

$$\int \sin^3(x) dx$$

Separate one factor of sine

$$\int \sin^2(x) \sin(x) dx$$

Apply the Pythagorean identity

$$\sin^2(x) = 1 - \cos^2(x)$$

Rewrite the integral

$$\int (1 - \cos^2(x)) \sin(x) dx$$

Let

$$u = \cos(x) \quad du = -\sin(x) dx$$

Substitute

$$-\int (1 - u^2) du$$

Integrate

$$\frac{u^3}{3} + u + C$$

Substitute back

$$\boxed{\frac{\cos^3(x)}{3} + \cos(x) + C}$$

c. Integration by parts

$$\int x \sin(2x) dx$$

Let

$$u = x \quad dv = \sin(2x) dx$$

Then

$$du = dx \quad v = -\frac{1}{2}\cos(2x)$$

Apply integration by parts

$$\int u dv = uv - \int v du$$

$$\int x \sin(2x) dx = -\frac{x}{2}\cos(2x) - \int -\frac{1}{2}\cos(2x) dx$$

Integrate

$$= -\frac{x}{2}\cos(2x) - \left(-\frac{1}{2}\right)\left(\frac{1}{2}\sin(2x)\right) + C$$

Simplify

$$\boxed{\frac{1}{4}\sin(2x) - \frac{x}{2}\cos(2x) + C}$$

Question 6

Three integrals are given that, although they look similar, may require different techniques of integration. Evaluate the integrals.

a. u -substitution

$$\int x \cos(x^2) dx$$

Let

$$u = x^2$$

$$du = 2x dx$$

$$x dx = \frac{1}{2} du$$

Substitute

$$\frac{1}{2} \int \cos(u) du$$

Integrate

$$\frac{1}{2} \sin(u) + C$$

Substitute back

$$\boxed{\frac{1}{2} \sin(x^2) + C}$$

b. Half-angle identity + integration by parts

$$\int x \cos^2(x) dx$$

Apply the half-angle identity

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

Rewrite the integral

$$\int x \left(\frac{1 + \cos(2x)}{2} \right) dx$$

Split the integral

$$\frac{1}{2} \int x dx + \frac{1}{2} \int x \cos(2x) dx$$

For

$$\int x \cos(2x) dx$$

Let

$$u = x \quad dv = \cos(2x) dx$$

$$du = dx \quad v = \frac{1}{2} \sin(2x)$$

Apply integration by parts

$$\int x \cos(2x) dx = \frac{x}{2} \sin(2x) - \int \frac{1}{2} \sin(2x) dx$$

Integrate

$$= \frac{x}{2} \sin(2x) - \frac{1}{2} \left(-\frac{1}{2} \cos(2x) \right)$$

Substitute back

$$\frac{x^2}{4} + \frac{1}{2} \left(\frac{x}{2} \sin(2x) - \frac{1}{2} \left(-\frac{1}{2} \cos(2x) \right) \right)$$

Simplify

$$\boxed{\frac{x^2}{4} + \frac{x}{4} \sin(2x) + \frac{1}{8} \cos(2x) + C}$$

c. Repeated integration by parts

$$\int x^2 \cos(x) dx$$

First integration by parts

$$u = x^2 \quad dv = \cos(x) dx$$

$$du = 2x dx \quad v = \sin(x)$$

Apply integration by parts

$$\int x^2 \cos(x) dx = x^2 \sin(x) - 2 \int x \sin(x) dx$$

For

$$\int x \sin(x) dx$$

Second integration by parts

$$u = x \quad dv = \sin(x) dx$$

$$du = dx \quad v = -\cos(x)$$

$$\int x \sin(x) dx = -x \cos(x) - \int -\cos(x) dx$$

$$= \sin(x) - x \cos(x)$$

Substitute back

$$x^2 \sin(x) - 2(\sin(x) - x \cos(x)) + C$$

Simplify

$$x^2 \sin(x) - 2 \sin(x) + 2x \cos(x) + C$$

Question 9

Evaluate the integral.

$$\int \frac{\cos(x)}{1 - \sin(x)} dx$$

Let

$$u = 1 - \sin(x)$$

Then

$$du = -\cos(x) dx$$

So

$$\cos(x) dx = -du$$

Substitute

$$-\int \frac{1}{u} du$$

Integrate

$$-\ln(1 - \sin(x)) + C$$

Question 11

Evaluate the integral.

$$\int_1^4 \sqrt{y} \ln y dy$$

Let

$$u = \ln y \quad dv = y^{\frac{1}{2}} dy$$

$$du = \frac{1}{y} dy \quad v = \frac{2}{3} y^{\frac{3}{2}}$$

Apply integration by parts

$$\int_1^4 \sqrt{y} \ln y \, dy = \frac{2}{3} y^{\frac{3}{2}} \ln y - \int \frac{2}{3} y^{\frac{3}{2}} \left(\frac{1}{y} \right) dy$$

Simplify

$$\frac{2}{3} y^{\frac{3}{2}} \ln y - \frac{2}{3} \int y^{\frac{1}{2}} dy$$

Integrate

$$\begin{aligned} &= \frac{2}{3} y^{\frac{3}{2}} \ln y - \frac{2}{3} \left(\frac{2}{3} y^{\frac{3}{2}} \right) \\ &= \frac{2}{3} y^{\frac{3}{2}} \ln y - \frac{4}{9} y^{\frac{3}{2}} \end{aligned}$$

Evaluate the upper bound

$$\begin{aligned} &= \frac{2}{3} (4)^{\frac{3}{2}} \ln 4 - \frac{4}{9} (4)^{\frac{3}{2}} \\ &\quad \frac{16}{3} \ln 4 - \frac{32}{9} \end{aligned}$$

Evaluate the lower bound

$$\begin{aligned} &= \frac{2}{3} (1)^{\frac{3}{2}} \ln 1 - \frac{4}{9} (1)^{\frac{3}{2}} \\ &\quad - \frac{4}{9} \end{aligned}$$

Subtract

$$\begin{aligned} &\frac{16}{3} \ln 4 - \frac{32}{9} - \left(-\frac{4}{9} \right) \\ &\frac{16}{3} \ln 4 - \frac{32}{9} + \frac{4}{9} \end{aligned}$$

$$\boxed{\frac{32}{3} \ln 2 - \frac{28}{9}}$$

Question 13

Evaluate the integral.

$$\int \frac{\ln(\ln y)}{y} dy$$

Let

$$u = \ln y$$

Then

$$du = \frac{1}{y} dy$$

Substitute

$$\int \ln(u) du$$

Apply integration by parts

Let

$$U = \ln(u) \quad dV = du$$

Then

$$dU = \frac{1}{u} du \quad V = u$$

Using

$$\int U dV = UV - \int V dU$$

we get

$$\int \ln(u) du = u \ln(u) - \int u \left(\frac{1}{u} \right) du$$

Integrate

$$u \ln(u) - u + C$$

Substitute back

$$\ln y [\ln(\ln y) - 1] + C$$

Question 14

Evaluate the integral.

$$\int_0^1 \frac{x}{(2x+1)^3} dx$$

Let

$$u = 2x + 1$$

$$du = 2 dx$$

$$dx = \frac{1}{2} du$$

Solve for x

$$x = \frac{u-1}{2}$$

Convert the bounds

$$x = 1 \Rightarrow u = 3$$

$$x = 0 \Rightarrow u = 1$$

Substitute

$$\begin{aligned} & \frac{1}{2} \int_1^3 \frac{u-1}{(u)^3} du \\ &= \frac{1}{4} \int_1^3 \frac{u-1}{(u)^3} du \end{aligned}$$

Rewrite the integral

$$\frac{1}{4} \int_1^3 (u^{-2} - u^{-3}) du$$

Integrate

$$\frac{1}{4} \left[-\frac{1}{u} + \frac{1}{2} \left(\frac{1}{u^2} \right) \right]_1^3$$

Evaluate the upper bound

$$-\frac{1}{3} + \frac{1}{2} \left(\frac{1}{3^2} \right)$$

$$-\frac{6}{18} + \frac{1}{18}$$

$$-\frac{5}{18}$$

Evaluate the lower bound

$$-\frac{1}{1} + \frac{1}{2} \left(\frac{1}{1^2} \right)$$

$$-\frac{2}{2} + \frac{1}{2}$$

$$-\frac{1}{2}$$

Subtract

$$\frac{1}{4} \left[-\frac{5}{18} - \left(-\frac{1}{2} \right) \right]$$

$$\frac{1}{4} \left(-\frac{5}{18} + \frac{9}{18} \right)$$

$$\frac{1}{4} \left(\frac{2}{9} \right)$$

$$\boxed{\frac{1}{18}}$$

Question 15

Evaluate the integral.

$$\int \frac{x}{x^4 + 9} dx$$

Let

$$u = x^2$$

$$du = 2x \, dx$$

$$x, dx = \frac{1}{2} du$$

Substitute

$$\frac{1}{2} \int \frac{1}{u^2 + 3^2} du$$

Apply the arctan form

$$\frac{1}{2} \left[\frac{1}{3} \arctan \left(\frac{u}{3} \right) \right] + C$$

Simplify

$$\frac{1}{6} \arctan \left(\frac{u}{3} \right) + C$$

Substitute back

$$\boxed{\frac{1}{6} \arctan \left(\frac{x^2}{3} \right) + C}$$

Question 17

Evaluate the integral.

$$\int_2^4 \frac{x + 2}{x^2 + 3x - 4} dx$$

Factor the denominator

$$x^2 + 3x - 4 = (x + 4)(x - 1)$$

Set up the decomposition

$$\frac{x + 2}{(x + 4)(x - 1)} = \frac{A}{x + 4} + \frac{B}{x - 1}$$

Clear the denominators

$$x + 2 = A(x - 1) + B(x + 4)$$

Let $x = 1$

$$1 + 2 = A(1 - 1) + B(1 + 4)$$

$$3 = 0 + 5B$$

$$B = \frac{3}{5}$$

Let $x = -4$

$$-4 + 2 = A(-4 - 1) + B(-4 + 4)$$

$$-2 = -5A + 0$$

$$A = \frac{2}{5}$$

Rewrite the integral

$$\int_2^4 \left(\frac{2}{5} \right) \frac{1}{x + 4} + \left(\frac{3}{5} \right) \frac{1}{x - 1}$$

Integrate

$$\left[\frac{2}{5} \ln |x + 4| + \frac{3}{5} \ln |x - 1| \right]_2^4$$

Evaluate the upper bound

$$\frac{2}{5} \ln 8 + \frac{3}{5} \ln 3$$

Evaluate the lower bound

$$\frac{2}{5} \ln 6$$

Subtract

$$\frac{2}{5} \ln 8 + \frac{3}{5} \ln 3 - \frac{2}{5} \ln 6$$

Simplify the logarithms

$$\frac{1}{5} (6 \ln 2 + 3 \ln 3 - 2 \ln 2 - 2 \ln 3)$$

$$\boxed{\frac{4}{5} \ln 2 + \frac{1}{5} \ln 3}$$

Question 24

Evaluate the integral.

$$\int_0^{\frac{\sqrt{2}}{2}} \frac{x^2}{\sqrt{1-x^2}} dx$$

Choose the trigonometric substitution

$$x = \sin(\theta) \quad dx = \cos(\theta) d\theta$$

Simplify the radical

$$\sqrt{1-x^2} = \sqrt{1-\sin^2(\theta)}$$

Recognize the Pythagorean identity $1 - \sin^2(\theta) = \cos^2(\theta)$

$$\sqrt{\cos^2(\theta)} = \cos(\theta)$$

Apply the trigonometric substitution

$$\int_0^{\frac{\sqrt{2}}{2}} \frac{\sin^2(\theta)}{\cos(\theta)} \cos(\theta) d\theta = \int_0^{\frac{\sqrt{2}}{2}} \sin^2(\theta) d\theta$$

Change the bounds

For $x = 0$

$$0 = \sin(\theta) \implies \theta = 0$$

For $x = \frac{\sqrt{2}}{2}$

$$\frac{\sqrt{2}}{2} = \sin(\theta) \implies \theta = \frac{\pi}{4}$$

Therefore

$$\int_0^{\frac{\pi}{4}} \sin^2(\theta) d\theta$$

Rewrite using the half-angle identity

$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

So

$$\frac{1}{2} \int_0^{\frac{\pi}{4}} (1 - \cos(2\theta)) d\theta$$

Integrate

$$\frac{1}{2} \left[\theta - \frac{1}{2} \sin(2\theta) \right]_0^{\frac{\pi}{4}}$$

Evaluate at the upper bound

$$\begin{aligned} & \frac{1}{2} \left(\frac{\pi}{4} - \frac{1}{2} \sin\left(\frac{\pi}{2}\right) \right) \\ &= \frac{1}{2} \left(\frac{\pi}{4} - \frac{1}{2} \right) \\ &= \frac{\pi}{8} - \frac{1}{4} \end{aligned}$$

Evaluate at the lower bound

$$\frac{1}{2} \left(0 - \frac{1}{2} \sin(0) \right) = 0$$

Subtract

$$\left(\frac{\pi}{8} - \frac{1}{4}\right) - 0$$

$$\boxed{\frac{\pi}{8} - \frac{1}{4}}$$

Question 32

Evaluate the integral.

$$\int (1 + \tan(x))^2 \sec(x) \, dx$$

Expand the squared expression

$$(1 + \tan(x))^2 = 1 + 2 \tan(x) + \tan^2(x)$$

So

$$\int (1 + 2 \tan(x) + \tan^2(x)) \sec(x) \, dx$$

Distribute $\sec(x)$

$$\int (\sec(x) + 2 \tan(x) \sec(x) + \tan^2(x) \sec(x)) \, dx$$

Apply the Pythagorean identity $\tan^2(x) = \sec^2(x) - 1$

$$\int (\sec(x) + 2 \tan(x) \sec(x) + \sec(x) (\sec^2(x) - 1)) \, dx$$

Distribute

$$\int (\sec(x) + 2 \tan(x) \sec(x) + \sec^3(x) - \sec(x)) \, dx$$

Simplify

$$\int (2 \tan(x) \sec(x) + \sec^3(x)) \, dx$$

Split the integral

$$2 \int \tan(x) \sec(x) \, dx + \int \sec^3(x) \, dx$$

Integrate the first term

$$2 \int \tan(x) \sec(x) dx = 2 \sec(x)$$

Rewrite the second term

$$\int \sec^3(x) dx = \sec^2(x) \sec(x) dx$$

Apply integration by parts

Let

$$u = \sec(x) \quad dv = \sec^2(x) dx$$

$$du = \sec(x) \tan(x) dx \quad v = \tan(x)$$

Apply

$$\int u dv = uv - \int v dv$$

$$\sec(x) \sec^2(x) dx = \sec(x) \tan(x) - \int \sec(x) \tan^2(x) dx$$

Apply the Pythagorean identity $\tan^2(x) = \sec^2(x) - 1$

$$\sec(x) \sec^2(x) dx = \sec(x) \tan(x) - \int \sec(x) (\sec^2(x) - 1) dx$$

Distribute

$$= \sec(x) \tan(x) - \int \sec^3(x) dx + \int \sec(x) dx$$

Add $\int \sec^3(x) dx$ to both sides

$$2 \int \sec^3(x) dx = \sec(x) \tan(x) + \int \sec(x) dx$$

Integrate

$$\int \sec(x) dx = \ln |\sec(x) + \tan(x)|$$

Divide by 2

$$\int \sec^3(x) dx = \frac{1}{2} \sec(x) \tan(x) + \frac{1}{2} \ln |\sec(x) + \tan(x)| + C$$

Substitute back into the original integral

$$\boxed{2 \sec(x) + \frac{1}{2} \sec(x) \tan(x) + \frac{1}{2} \ln |\sec(x) + \tan(x)| + C}$$

Question 37

Evaluate the integral.

$$\int \ln(x + \sqrt{x^2 - 1}) dx$$

Let

$$u = \ln(x + \sqrt{x^2 - 1}) \quad dv = dx$$

$$v = x$$

Differentiate u

$$du = \frac{1 + \frac{x}{\sqrt{x^2 - 1}}}{x + \sqrt{x^2 - 1}} dx$$

Simplify du

$$1 + \frac{x}{\sqrt{x^2 - 1}} = \frac{x + \sqrt{x^2 - 1}}{\sqrt{x^2 - 1}}$$

so

$$du = \frac{1}{\sqrt{x^2 - 1}} dx$$

Apply integration by parts

$$\int u dv = uv - \int v du$$

$$x \ln(x + \sqrt{x^2 - 1}) - \int \frac{x}{\sqrt{x^2 - 1}} dx$$

Apply u -substitution to the remaining integral

Let

$$w = x^2 - 1$$

$$dw = 2x \, dx$$

$$x \, dx = \frac{1}{2} \, dw$$

Substitute

$$\int \frac{x}{\sqrt{x^2 - 1}} \, dx = \frac{1}{2} \int w^{-\frac{1}{2}} \, dw$$

Integrate

$$\frac{1}{2} \left(2w^{\frac{1}{2}} \right) = w^{\frac{1}{2}}$$

Substitute back $x^2 - 1$

$$x \ln \left(x + \sqrt{x^2 - 1} \right) - \sqrt{x^2 - 1} + C$$

Question 50

Evaluate the integral.

$$\int \frac{1}{x\sqrt{x-1}} \, dx$$

Apply substitution

Let

$$u = \sqrt{x-1}$$

Square both sides

$$u^2 = x - 1$$

So

$$x = u^2 + 1$$

Differentiate

$$dx = 2u \, du$$

Rewrite the integral in terms of u

$$\int \frac{2u \, du}{(u^2 + 1)u}$$

Cancel u

$$\int \frac{2}{u^2 + 1} \, du$$

Recognize the arctan form

$$2 \arctan(u) = 2 \arctan(u) + C$$

Substitute back

$2 \arctan(\sqrt{x-1}) + C$
